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A Human Evaluation

The evaluation proposed by Stanovsky et al. (2019) is automatic and thus bound to introduce some noise. To mitigate this concern, we perform a human evaluation which complements the automatic metric and provides a more thorough analysis of the results. The evaluation was applied to translations into Hebrew which were debiased according to the *1-token-profession* words (see Section 3), when specifically applied on encoder's input. We only analyze the sentences that changed in any way compared to the original (non debiased) translations. There were 184 out of 1584 sentences which changed (11%). 32% of the changed sentences improved the gender prediction (i.e the gender of a profession was incorrect in the original translation and correct in the debiased translation), 3% of the changed sentences were detracted (i.e the gender of a profession was correct in the original translation and incorrect in the debiased translation), and in the rest 65% the change did not affect the gender of the profession. In 23% out of the 65% of unaffected sentences, the gender of the other profession in the sentence was changed, for example in the sentence "The salesperson talked to the auditor because she was worried about the audit" the gender of salesperson did not change in the debiased translations, but the gender of the auditor was changed from male to female form. Note that this is not a mistake since the gender of auditor is unknown in the source sentence. This shows that the debiasing method affects a larger amount of professions which are not counted in the improvement of the model.

B Human Annotations

To define the gender direction in the target language for both debias methods, we needed the translations of the 10 representative gender word pairs for each language. To get those pairs, we asked a native speaker of each of these languages to translate them into their language. In the case of a pair that is irrelevant to the target language (like Mary and John which are common male and female names in English but not in other languages), we asked them to adapt the pair to represent gender pairs in their language. The set of professions that we debias was also translated into the target languages by three native speakers in each language. The professions annotations were taken from Iluz et al. (2023). The translations of the 10 pairs were collected for four languages, German, Hebrew, Russian, and Spanish. [7]

[7] link for the 10 pairs datasets will be released upon publication.

C Statistical Significance

In order to determine the statistical significance of our findings, we employed McNemar's test, as recommended by Dror et al. (2018). McNemar's test is designed for models with binary labels, therefore it is suitable to test the gender bias scores where each sentence is classified as correct if the gender is accurately identified in the translation and incorrect otherwise. The null hypothesis for this test states that the marginal probability for each outcome is equal between the two algorithms being compared, indicating that the models are identical. In our case, the two models being compared are the original translation model and the debiased version. When concatenating results per debias method, we get that the results of Hard Debias are significant with p-value of 3.01E-07, and the results of INLP are significant with p-value of 9.65E-06. When comparing the results per embedding table to debias, we get that debiasing the encoder inputs is significant with p-value of 5.42E-10, debiasing the decoder inputs is significant with p-value of 0.016 and debiasing the decoder outputs is significant with p-value of 0.014. finally when concatenating all the results, we get that comparing the outputs of a debiased model to a the original model, the results are significant with p-value of 0.01.